

1D-Array

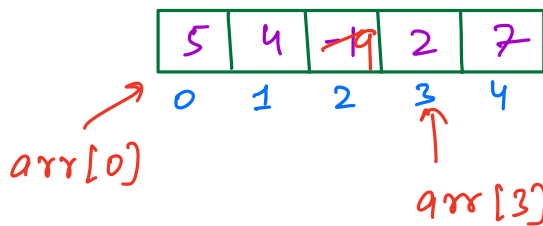
Array → sequential collection of similar data.

Train , YT playlist

Syntax

int arr[] = new int [5];
↓ ↓ ↓
type variable size of
 name array

int [] arr = new int [5];



arr[2] = 9

NO NEGATIVE INDEXING

int[] arr = {1, 2, 3};

Maximum index of array of size N ?
 $= N-1$

we know index starts from 0.

So for N -size array, indices will be :

0, 1, 2, 3, , $N-1$

Sum of all elements in array.

arr = { 3 4 1 5 1 }

$$ans = 3 + 4 + 1 + 5 + 1 = 14$$

Question 1

Given array of size N as input, print its sum.

find max element in array.

{ 3, 4, 1, 5, 1 }

find max b/w a & b.

```
if (a > b)    ans = a
else         ans = b
```

find max b/w a, b & c

ans = 0

```
if (a > ans)  ans = a;
if (b > ans)  ans = b;
if (c > ans)  ans = c;
```

Question 4

Given an integer array of size N as input,
check whether an integer K is present in array
or not.

arr = { 3, 8, 1, -10, 0 }

K = 8 → true

K = 3 → true

K = 2 → false

```

boolean findK ( int arr[], int n, int K ) {
    for ( int i = 0 ; i < n ; i++ ) {
        if ( arr[i] == K ) {
            return true;
        }
    }
    return false;
}

```

arr = { 3, 4, 1, 5, 1 }

ans = 2

find frequency of 1 ?

↳ count of occurrences of 1

Question 5

Given an integer array of size N as input,
find the frequency of K in array.

arr = { 1, 2, 1, 1, 3, 0, 2, 1, 2 }

K = 1 → 4

K = 5 → 0

```

int frequencyK ( int arr[], int n, int k ) {
    int count = 0;
    for (int i = 0; i < n; i++) {
        if ( arr[i] == k ) {
            count++;
        }
    }
    return count;
}

```

Question 6

Given an integer array as input, return the frequency count of array.

arr = { 1, 1, 2, 1, 3, 1, 3 }

output = { 4, 4, 1, 4, 2, 4, 2 }

```
int[] freqCount (int arr[]) {
```

```
    int n = arr.length;
```

```
    int ans[] = new int[n];
```

```
    for (int i=0; i<n; i++) {
```

```
        ans[i] = frequencyK (arr, n, arr[i])
```

```
    } ↳ called N times
```

```
    return ans;
```

```
}
```

Question 7

Given an integer array as input,

check whether it is strictly increasing.

arr = { 0, 2, 6, 7, 8 } ⇒ true

arr = { 0, 2, 2, 6, 7, 7 } ⇒ false

arr = { 0, 5, 2, 5, 10 } ⇒ false

```
boolean isStrictlyInc (int arr[]) {
```

```
    int n = arr.length;
```

```
    for (int i=0; i+1 < n  
               OR  
               i < n-1; i++) {
```

```
        if (arr[i] >= arr[i+1]) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    return true;
```

```
}
```

$i = 0 \dots n-1$

$arr[0] < arr[1]$

$arr[1] < arr[2]$

$arr[2] < arr[3]$

$\vdots$

$arr[i] < arr[i+1]$

DOUBT

```
arr = new int[n];
```

```
for (i=0; i<n; i++) {
```

```
    arr[i] = sc.nextInt();
```

```
}
```

Notes link : [https://www.scaler.com/topics/java/online-java-compiler/?
snippet_slug=fdb676b211a2ec2a540e](https://www.scaler.com/topics/java/online-java-compiler/?snippet_slug=fdb676b211a2ec2a540e)